SQL Test 1

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1.Display the employee name and department name for employees who have a matching department using INNER JOIN.

```
select e.emp_name , d.dept_name from employees_1 as e

inner join departments as d

on e.Dept_ID = d.Dept_ID;
```

#2. Display all employees along with their department names. Employees without a matching department should also be displayed.

```
select e.Emp_ID, e.Salary, e.emp_name , d.dept_name from employees_1 as e

left join departments as d

on e.Dept_ID = d.Dept_ID;
```

#3. Display all departments along with the employees working in each department.

```
select   d.*,e.* from departments as d

left join employees_1 as e

on e.Dept_ID = d.Dept_ID;
```

4.Find the employees who do not have a matching department.

```
select  e.*, d.* from departments as d

right join employees_1 as e on e.Dept_ID = d.Dept_ID

where d.Dept_ID is null;
```

5.Find the departments that do not have any matching employees.

```
select d.* ,e.* from departments d

left join employees_1 e

on d.Dept_ID = e.Dept_ID where e.Emp_ID is null;
```

6.Display Emp_ID, Emp_Name, Salary, and Dept_Name for all employees who have a matching department.

```
select e.Emp_ID, e.Emp_Name, e.Salary, d.dept_name from employees_1 as e

inner join departments as d

on e.Dept_ID = d.Dept_ID;
```

7. Display all employees and their assigned project names, including employees who are not assigned to any project.

```
select e.*, p.* from employees_1 e

left join projects p

on p.Dept_ID = e.Dept_ID;
```

8.Display all projects along with the employee details associated with their department.

```
select p.*, e.*, d.Dept_Name from projects p

left join employees_1 e

on p.Dept_ID = e.Dept_ID

left join departments d

on d.Dept_ID = e.Dept_ID;
```

9.Display Emp_Name, Dept_Name, and Project_Name by joining all three tables.

```
select e.emp_name,d.Dept_Name, p.Project_Name from employees_1 e
```

inner join projects as p on e.Dept_ID = p.Dept_ID

inner join departments as d on p.Dept_ID = d.Dept_ID;

10.Find employees who belong to a department but do not have a project associated with their department.

select e.*, d.*, p.* from employees_1 e

left join departments d on d.Dept_ID = e.Dept_ID

left join projects p on p.Dept_ID = e.Dept_ID where p.Project_ID is null;

11.Display all departments and their corresponding project names, including departments that do not have any project.

select d.*, p.* from departments d

right join projects p on p.Dept_ID = d.Dept_ID;

12.Using INNER JOIN, display employees who belong to the IT or HR department.

select e.*, d.Dept_Name from employees_1 e

inner join departments d on d.Dept_ID = e.Dept_ID

where d.Dept_Name in ("HR" , "IT");

13.Using LEFT JOIN, display employees whose department information is missing.

select e.*, d.Dept_Name from employees_1 e

left join departments d on d.Dept_ID = e.Dept_ID

where d.Dept_ID is null;

14.Using RIGHT JOIN, display departments that have no employees.

```
select d.* , e.emp_name from employees_1 e

right join departments d on d.Dept_ID= e.Dept_ID

where e.Emp_ID is null;
```

15.Using FULL OUTER JOIN, display all employees and all departments, including both matching and non-matching records.

```
(select e.* , d.* from employees_1 e

left join departments d on d.Dept_ID = e.Dept_ID )

union

(select e.* , d.* from employees_1 e

right join departments d on d.Dept_ID = e.Dept_ID )
```